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$$\begin{aligned}\lim_{x \rightarrow 7} \frac{(3 - \sqrt{x+2})}{(x-7)} &= \frac{0}{0} \\&= \lim_{x \rightarrow \infty} \frac{(3 - \sqrt{x+2})(3 + \sqrt{x+2})}{(x-7)(3 + \sqrt{x+2})} \\&= \lim_{x \rightarrow \infty} \frac{(9 - (x+2))}{(x-7)(3 + \sqrt{x+2})} \\&= \lim_{x \rightarrow \infty} \frac{-(x-7)}{(x-7)(3 + \sqrt{x+2})} \\&= \lim_{x \rightarrow \infty} \frac{-1}{(3 + \sqrt{x+2})} \\&= \frac{-1}{(3+3)} \\&= \frac{-1}{6}\end{aligned}$$

References